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Glucocorticoid-Induced Cataract in Chick Embryo Monitored by Raman Spectroscopy

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Glucocorticoid-induced cataract lens in chick embryo was monitored by laser Raman spectroscopy. The lens opacity that appeared in chick embryo is a reversible one. Raman spectra show no significant change in the relative content of water or secondary structure of the proteins upon lens opacification. The intensity ratios of tyrosine doublet bands in Raman spectra between clear and opaque lens portions are changed. This change is reversible, and is interpreted as a protein-water phase separation that occurred during lens opacification. Invest Ophthalmol Vis Sci 30:132–137, 1989

Glucocorticoid hormones are physiologically important steroid hormones, however, their actions are multiple and diverse on many organs. Cataract, usually appearing as a posterior subcapsular cataract, is one of the side effects caused by corticoid treatment in humans. The mechanism of glucocorticoid-induced cataract formation is still definitely unknown.[1] Newly synthesized steroid hormones have been developed and used systematically or topically in many patients for a long time. The incidence of steroid-induced cataract has increased among these patients. As a direct effect, glucocorticoid could alter the metabolism in the lens and also produce lens opacity.[2–6] Nishigori, Lee and Iwatsuru succeeded in inducing a reversible cataract on the lens of developing chick embryo within a short period after a single administration of corticoid.[7–9] This cataract is not directly related to human cataract induced by steroid; however, it provides some valuable information on glucocorticoid action on the lens. Cold cataract is also a reversible phenomenon that we have monitored by laser Raman spectroscopy.[10]

Laser Raman spectroscopy is one of the most useful techniques for nondestructively monitoring a lens in situ and provides information at the molecular level.[11–18] We have reported a reversible protein tyrosine microenvironmental change monitored by Raman spectroscopy in cold cataract lens.[10] In this study we measured the Raman spectra of this particular corticoid-induced cataract in chick embryo lens.

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Materials and Methods

Day 1 fertile White Leghorn eggs were obtained from a commercial hatchery and kept in an incubator at 37.5°C and 68% relative humidity. Chick embryos of day 15 in ovo were used and administered 0.25 μmol/egg of sodium hydrocortisone hemisuccinate (HC) injected directly into the air chamber through a pinhole in the egg shell. The pinhole was sealed with cellophane tape. The eggs were then kept in an incubator at same conditions for 48 hr and 96 hr. The lenses developed opacity at various stage after 48 hr of treatment.[19] The lenses with stage V (nuclear opacity) or III (ring opacity) cataract[7] were carefully excised and subjected to Raman spectral measurement. Care of animals conformed to the ARVO Resolution on the Use of Animals in Research.

The lens excised was put into a cuvette cell filled with Hank's MEM solution and placed horizontally on the cell bottom. The incident laser beam illuminated from the cell bottom was focused at the nuclear center of the lens and the illuminated region was changed horizontally by moving the cuvette cell itself with a microscale manipulator. Raman scattering signals were collected at 90° of the incident laser beam from a microscopic cylindical column, approximately 0.49 μl of volume, at several different points moving every 0.4 mm along the horizontal axis of the lens.[18,20] Raman spectra were measured by using a JASCO NR-1000 Raman spectrometer equipped with a photomultiplier tube (Hamamatsu Photonics, R-464, Shizuoka, Japan). The excitation wavelength employed was 514.5 nm from an argon laser (NEC, GLG 3200, Tokyo, Japan) and the laser power was about 80 mW at the sample position. Peak frequen-

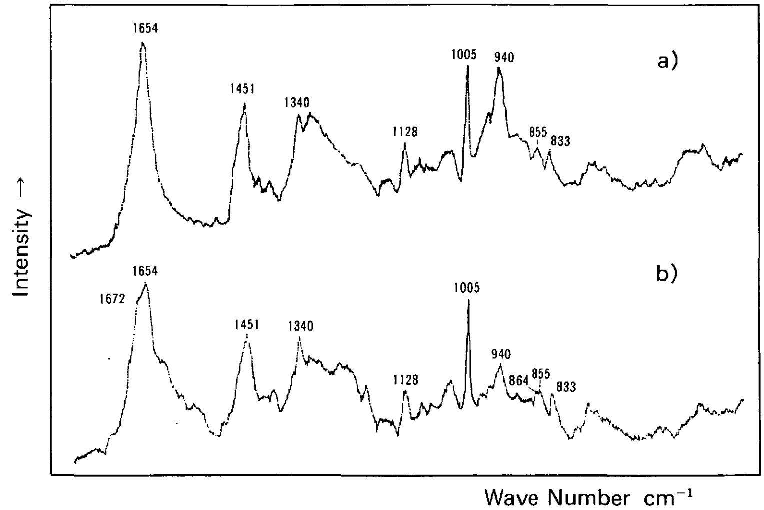

Fig. 1. Raman spectra from the 17-day in ovo normal control lenses at spectral region from 450 cm^{-1} to 1800 cm^{-1}; (**a**) nuclear center, (**b**) 0.8 mm from the center.

cies were calibrated by measuring the spectrum of indene.

Results

Normal Control Lens

Raman spectra from the nuclear center and the cortex (0.8 mm from the center) of normal chick embryo lens at day 17 in ovo is shown in Figure 1 at the spectral region from 450 cm^{-1} to 1800 cm^{-1}. The Raman spectrum of the chick embryo lens represents mainly spectral bands of lens proteins and is generally similar to that of the hen lens.[21] The difference between these Raman spectra at nucleus and cortex reveals the difference in both amino acid composition and conformation in these regions. Main Raman bands can be assigned to protein molecular vibration, for example, the Raman lines at 1654 and 940 cm^{-1} indicate α-helical conformation and Amide III at 1240 cm^{-1} indicates β-structure. A peak at 1005 cm^{-1} is assigned as phenylalanine. At the nuclear center of the lens, a band at 940 cm^{-1}, indicating α-helical conformation of the nuclear proteins, is mainly composed of δ-crystallin, a unique bird lens protein. The C-N streching vibration of the protein backbone at 1128 cm^{-1} was used as an internal reference for the spectra. The ratio of the intensity of the bands ($I_{940}/I_{1128}cm^{-1}$) at the various points of the lens indicates lower concentration of δ-crystallin in the cortex nearer the equator (Fig. 1b). In the region near the cortex equator, a slight increase of two bands is seen at 1672 and 1240 cm^{-1}, indicating the existence of protein β-structure.[21,22]

Figure 2a shows Raman spectra in the 740–900 cm^{-1} region from 0.8 mm portions of the day 17 in ovo chick lenses. Two peaks at 833 and 855 cm^{-1} are assigned as tyrosine doublet.[23-26] The intensity ratio of tyrosine doublet (I_{833}/I_{855}) is 1.63 ± 0.01 and 1.29 ± 0.01 at the nuclear center and at 0.8 mm portions, respectively, in the day 17 in ovo lenses. In day 19 in ovo lenses the ratios are 1.69 ± 0.02 and 1.71 ± 0.05 at the nuclear center and 0.8 mm portions, respectively (Fig. 3).

Complete absence of S-S bonds at 509 cm^{-1} in lens nucleus indicates a δ-crystallin nature. At the same time, the absence of a protein sulfhydryl group, which usually appeared at the 2580 cm^{-1} region, is also observed in the lens nucleus.[26-28]

Intensity ratio of OH streching mode at 3390 cm^{-1} and CH streching mode at 2935 cm^{-1} is a good indicator for lens dehydration and hydration.[29] In chick embryo lens the relative concentration of water does not change significantly as a function of location (Fig. 4).

Glucocorticoid-Induced Cataract Lens

Relative water contents of the lenses with stage III and V cataract at various regions indicate no significant change compared with those of normal lenses (Fig. 4). In this type of cataract, lens hydration does not occur even at the opaque site where many vacuoles are observed histologically.[30]

In the Raman spectral region from 450 cm^{-1} to 1800 cm^{-1}, there is no significant alteration upon lens opacification in protein backbone amino acid composition or protein conformation, except in the case of tyrosine residues (Fig. 5). The ratio of I_{940}/I_{1128}

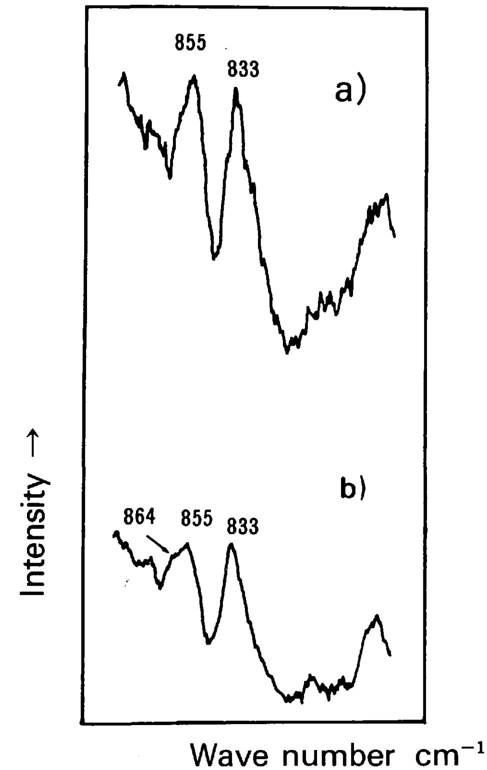

Fig. 2. Raman spectra from the 17-day in ovo lenses at 0.8 mm from the center; (**a**) normal control clear portion, (**b**) corticoid-treated opaque portion (spectral region from 740 cm^{-1} to 900 cm^{-1}).

indicates a similar content of α-helical conformations at the nuclear center, and 0.4 mm and 0.8 mm from the center, between the normal control lens and stage III cataract lens with ring opacity. There is little change in β-structure at the 1672 cm^{-1} or 1240 cm^{-1} Raman lines at three locations in the cataractous lens. These results indicate that the secondary structure of the lens proteins does not change significantly upon lens opacification.

The intensity ratio of the tyrosine doublet (I_{833}/I_{855}) has been shown to be a good marker for lens opacification.[10,23-25] In the chick embryo the peaks of the tyrosine doublet appear at 833 cm^{-1} and 855 cm^{-1} even in opaque lens (Fig. 2b). In the stage III steroid-induced cataract, the clear nuclear center reveals no significant change in the intensity ratio of the tyrosine doublet in the day 17 in ovo chick after 48 hr of corticoid treatment. In the opaque portion on the ring opacity, 0.8 mm from the nuclear center along the horizontal axis, the intensity ratio of the tyrosine doublet is altered from 1.29 ± 0.01 of the normal control to 1.80 ± 0.09 of the ring opacity lens (Fig. 3). This result means that some tyrosine residues of lens proteins change microenvironmentally from hydrophylic to hydrophobic upon lens opacification in the lens cortex with ring cataract.[10,30] In this situation the tyrosine residues are thought to become hydrophobic and are buried between aggregated proteins upon lens opacification. In the stage V cataract lens, a similar microenvironmental change of tyrosine residues is detected at 0.8 mm from the nuclear center. However, we cannot measure the Raman spectrum of the tyrosine doublet at the opaque nuclear center because the laser beam does not penetrate well.

The lens opacity at various stages after 48 hr of HC treatment is reversible to lower stage opacity after 96 hr when, 75% of lenses appear to be transparent by microscopic examination.[7] The values of I_{833}/I_{855} in normal control lenses alter from 1.29 at day 17 in ovo (48 hr after corticoid treatment) to 1.74 at day 19 (96 hr) as a developmental change. The microenvironmental change in the tyrosine residues is also reversible in the transparent lens after 96 hr of HC treatment (Fig. 3); this takes into account developmental changes in normal control lenses from day 17 to 19 in ovo. The value of the intensity ratio of the tyrosine doublet is 1.60 ± 0.08, which is almost the same value as in normal control lens (1.74 ± 0.09). The microenvironmental change in tyrosine residues is thought to be a reversible one, depending on lens opacification.

Discussion

Glucocorticoid, one of the steroid hormones, has numerous actions on the living cells and tissues. Cataract is one of the undesirable side effects after long-term steroid therapy. Nishigori and his coworkers have developed a new animal model of steroid-induced cataract in chick embryo with a single injection of glucocorticoid. This lens opacity appears within 24 hr and reaches maximum opacity at various stages at 48 hr. Then the opacity disappears approximately within 96 hr after steroid injection.

At 48 hr after glucocorticoid injection, that is, day 17 in ovo, the lens opacities can be classified into two typical groups: stage III lens, with a ring opacity between the cortical region and the nucleus, and stage V lens, with an opaque nucleus expanded to the same region of ring opacity as in stage III. All of the glucocorticoid-treated lenses develop lens opacity above stage II at 48 hr. After 96 hr all lens opacities disappear in the nucleus and the ring opacity region.[7]

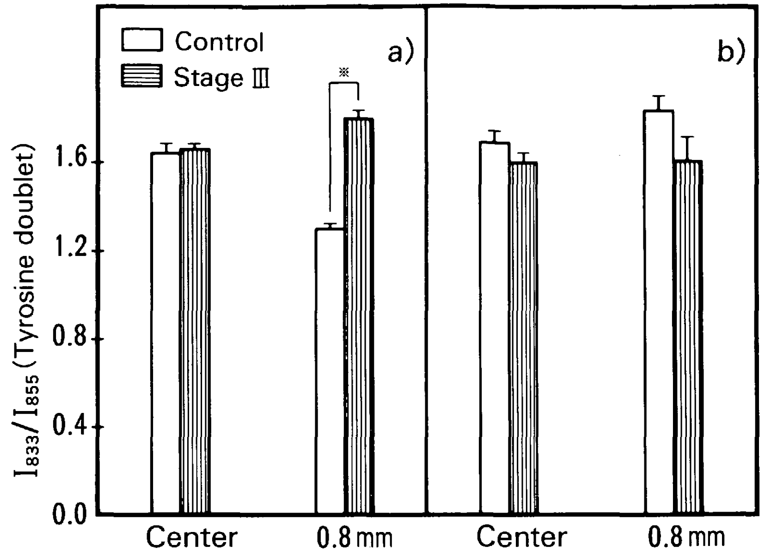

Fig. 3. Intensity ratio of tyrosin doublet at nucleus center and 0.8 mm from the center; **(a)** day 17 lens, **(b)** day 19 lens (means ± SD, n = 4, *$P < 0.005$).

Increase of the relative water content monitored by laser Raman spectroscopy in many other experimental animal cataractous lenses has been reported.[18,24,29] There is no significant change in relative water content in various portions from the nucleus center to cortex compared to normal control lenses and opaque lenses with stage III or V (Fig. 4). In this glucocorticoid-induced cataract, changes in lens water content do not play an important role.

Yu and his collegues demonstrated the Raman spectrum of the 10-month-old hen.[21] In the 700–900 cm^{-1} region the peak originating from tyrosine resi-

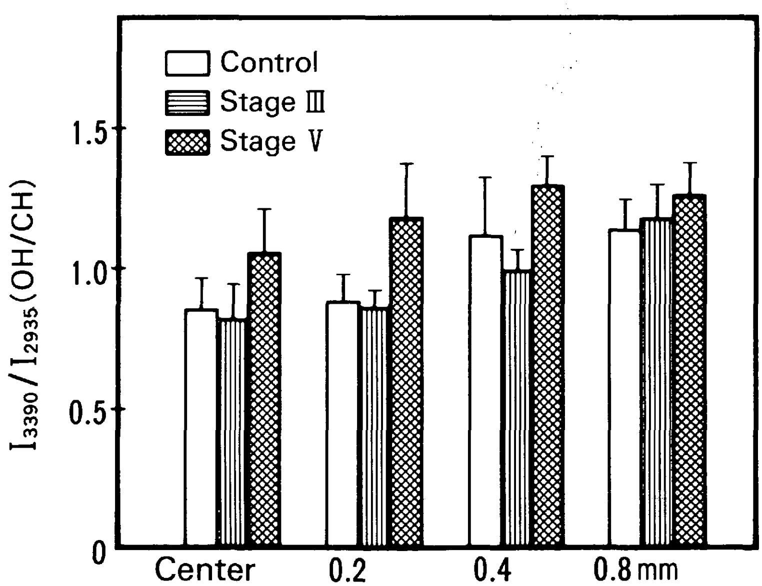

Fig. 4. Relative water contents detected by Raman spectroscopy at various portions and stages (means ± SD, n = 3–4).

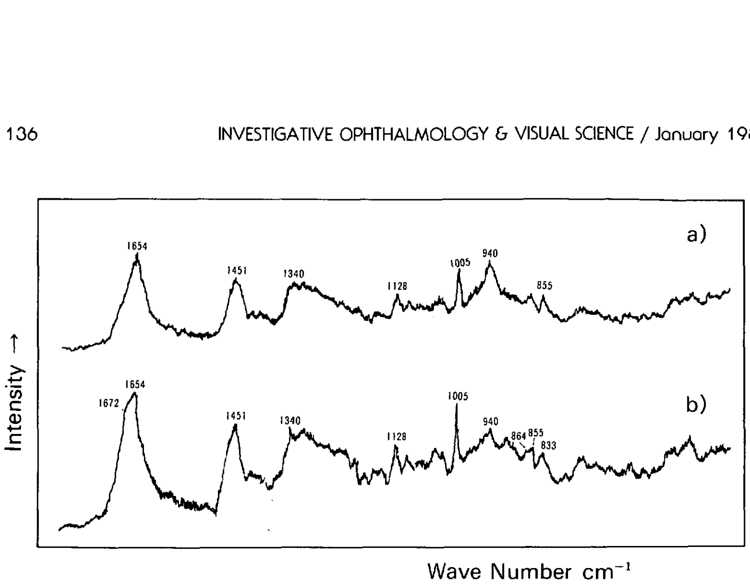

Fig. 5. Raman spectra from the day 17 in ovo stage III lenses (ring opacity) at spectral region from 450 cm^{-1} to 1800 cm^{-1}; (**a**) nuclear center (clear), (**b**) 0.8 mm from the center (opaque).

dues was assigned at 833 and 864 cm^{-1} in the spectrum of the hen lens. However, we show two bands at 833 and 855 cm^{-1} in the spectrum of day 17 in ovo chick lens. Figure 2 shows that the main bands of the tyrosine doublet appear at 833 and 855 cm^{-1} and the small peak or shoulder at 864 cm^{-1} also exists at this embryonal period. The peak at 864 cm^{-1} from tyrosine appears in the developed lens.[21]

A significant change in the Raman band upon lens opacification was observed in the 700–900 cm^{-1} region of the Raman spectrum. Using a stage III opaque lens at 48 hr, we demonstrated that only the opaque portion of the ring opacity shows a microenvironmental change in the tyrosine residues in the lens proteins. This change in intensity ratio (I_{833}/I_{855}) from 1.29 to 1.80 means that some tyrosine residues become hydrophobic upon opacity formation.[10] This change was observed uniformly in the case of cataract formation.[10,17,24,29] Especially in cold cataract, which is well known to be a reversible opacity formation, the microenvironmental change in the tyrosine residues is reversible, depending on the temperature.[10] The phenomenon of the cold cataract is interpreted as a water-protein phase separation, that is, it is thought that the phase separation of water and proteins occurs when the lens become opaque.[31-36] At day 19 in ovo after 96 hr of glucocorticoid treatment, the intensity ratios of the tyrosine doublet of the normal control and treated lenses are almost the same (detected at 0.8 mm from the nucleus), though the values of I_{833}/I_{855} alter from 1.29 at day 17 in ovo to 1.74 at day 19 in ovo in normal control lenses as a developmental change from day 17 in ovo to day 19. The intensity ratio at 19 days in ovo, treated lens, which reverse to transparency, is 1.60 at the 0.8 mm portion. Therefore the microenvironmental change in tyrosine is thought to be a reversible one, taking into account developmental change. Moreover, we can posit that the lens opacity formation in the glucocorticoid-treated chick embryo is due to a water-protein phase separation.

In glucocorticoid-induced cataractous lenses, the following phenomena are common: an elevation of glucose level,[19] a decline of glutathion content[7,8] and an increase of lipid peroxide.[37] It is uncertain whether these phenomena are related to the phase separation of water and protein in the lens. Therefore, further investigations are necessary to elucidate the water-protein phase separation mechanism of lens opacification in its relation to biochemical changes.

Key words: chick embryo, Raman spectroscopy, steroid cataract, tyrosine residues, protein-water phase separation

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